

A Practical Guide to Mean squared error

TOPIC -- MEAN SQUARED ERROR

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

-- 01 --

$MSE(S_{n-1}^2) = \frac{1}{n} (\mu^4 - n - 3n - 1 \sigma^4) = \frac{1}{n} (\gamma^2 + 2n - 1) \sigma^4$,

$$\{MSE\} (S_{n-1}^2) = \frac{1}{n} \left(\mu^4 - \frac{n-3}{n-1} \sigma^4 \right) = \frac{1}{n} \left(\gamma^2 + \frac{2n-1}{n-1} \sigma^4 \right)$$

-- 02 --

Mean Suppose we have a random sample of size n from a population, $X_1, \dots, X_n$. Suppose the sample units were chosen with replacement. That is, the n units are selected one at a time, and previously selected units are still eligible for selection for all n draws. The usual estimator for the population mean μ is the sample average

-- 03 --

For a Gaussian distribution, where $\gamma^2 = 0$, this means that the MSE is minimized when dividing the sum by $a = n + 1$. The minimum excess kurtosis is $\gamma^2 = -2$, which is achieved by a Bernoulli distribution with $p = 1/2$ (a coin flip), and the MSE is minimized for $a = n - 1 + 2/n$. Hence regardless of the kurtosis, we get a "better" estimate (in the sense of having a lower MSE) by scaling down the unbiased estimator a little bit; this is a simple example of a shrinkage estimator: one "shrinks" the estimator towards zero (scales down the unbiased estimator). Further, while the corrected sample variance is the best unbiased estimator (minimum mean squared error among unbiased estimators) of variance for Gaussian distributions, if the distribution is not Gaussian, then even among unbiased estimators, the best unbiased estimator of the variance may not be S_{n-1}^2 .

-- 04 --

An even shorter proof can be achieved using the well-known formula that for a random variable X , $E(X^2) = \text{Var}(X) + (E(X))^2$. By substituting X with $\hat{\theta} - \theta$, we have

-- 05 --

But in real modeling case, MSE could be described as the addition of model variance, model bias, and irreducible uncertainty (see Bias–variance tradeoff). According to the relationship, the MSE of the estimators could be simply used for the efficiency comparison, which includes the information of estimator variance and bias. This is called MSE criterion.

-- 06 --

where μ_4 is the fourth central moment of the distribution or population, and $\gamma_2 = \mu_4 / \sigma^4 - 3$ is the excess kurtosis. However, one can use other estimators for σ^2 which are proportional to S_{n-1}^2 , and an appropriate choice can always give a lower mean squared error. If we define

-- 07 --

In other words, the MSE is the mean $\left(\frac{1}{n} \sum_{i=1}^n \right)$ of the squares of the errors $(Y_i - \hat{Y}_i)^2$. This is an easily computable quantity for a particular sample (and hence is sample-dependent). In matrix notation,

-- 08 --

Criticism The use of mean squared error without question has been criticized by the decision theorist James Berger. Mean squared error is the negative of the expected value of one specific utility function, the quadratic utility function, which may not be the appropriate utility function to use under a given set of circumstances. There are, however, some scenarios where mean squared error can serve as a good approximation to a loss function occurring naturally in an application. Like variance, mean squared error has the disadvantage of heavily weighting outliers. This is a result of the squaring of each term, which effectively weights large errors more heavily than small ones. This property, undesirable in many applications, has led researchers to use alternatives such as the mean absolute error, or those based on the median.

$$\begin{aligned} \text{MSE} \blacksquare (S_{n-2}) &= E \blacksquare [(n-1)aS_{n-12} - \sigma^2] = E \blacksquare [(n-1)2a2S_{n-14} - 2(n-1)aS_{n-12})\sigma^2 + \sigma^4] = (n-1)2a2E \blacksquare [S_{n-14}] - 2(n-1)aE \blacksquare [S_{n-12})\sigma^2 + \sigma^4] = (n-1)2a2E \blacksquare [S_{n-14}] - 2(n-1)a\sigma^4 + \sigma^4E \blacksquare [S_{n-12}] = \sigma^2 = (n-1)2a2(\gamma^2n + n + 1n-1)\sigma^4 - 2(n-1a)\sigma^4 + \sigma^4E \blacksquare [S_{n-14}] = \text{MSE} \blacksquare (S_{n-12}) + \sigma^4 = n-1na2((n-1)\gamma^2 + n^2 + n)\sigma^4 - 2(n-1a)\sigma^4 + \sigma^4 \end{aligned}$$
$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta}, \theta)^2.$$